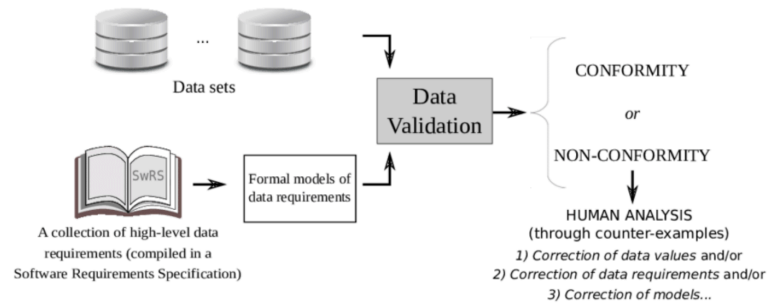


Data Validation Tool

Created by Kumar, Shobhit, last modified on Jun 30, 2022

Overview

Data validation means checking the accuracy and quality of target data when compared to source data while migrating the data. Different types of validation can be performed depending on destination constraints or objectives. Data validation is a form of building confidence in Data migrations.



Objective

1. To ensure the data is accurate and complete
2. To ensure the quality of data.
3. As we are migrating from Oracle to PostgreSQL, the data type may vary. Data validation will help us identify any such issues before hand.

Design Principles

1. Data type Validation

Data type validation is customarily carried out on one or more simple data fields.

The simplest kind of data type validation verifies that the individual characters provided through user input are consistent with the expected characters of one or more known primitive data types as defined in a programming language or data storage and retrieval mechanism.

For example, an integer field may require input to use only characters 0 through 9.

2. Data correctness

Data should be matching with the Source data when compared.

When we compare the data it should be exactly in the same form as it is in Source.

3. Data count

Row count should be same in both source and Target.

Count of rows in source can be more sometimes if there is a lag in replication but should not be the case other way round.

4. Mismatch reporting

The tool should be able to generate the report having records not matching and with the attributes those are not matching.

5. Missing Report

The tool should be able to generate the report having records missing at target.

Different Tools for Data validation

There are many tools available for data validation/Comparison. Few of which are :

<https://wiki.intuit.com/pages/viewpage.action?spaceKey=EMSPD&title=Data+Validation+Tool>

Tool	Advantages	
Veridata	- Works well with - We are already using it for Oracle to Oracle compare today and it scales for the PSP Data volume - Works best with Oracle as it is a native tool. - Generates OOS report and repair script	- values than expected - For database with TIME data type that supports value outside range of 00:00:00 to 23:59:59, the compare will only work when Source and Target are the same database. Doesn't support PostgreSQL data compare Separate setup needed which included a server and DB.
DMS Validator	- DMS Validator is a out of box solution from AWS - Works well with smaller datasets/ where change volume is not very high - Can validate Heterogeneous databases i.e. when the source and target are different	- Data validation requires that the table has a primary key or unique index - Data validation isn't supported if a migration uses customised filtering or when consolidating several databases into one DMS validation doesn't scale for busy tables. Can't compare time range partition tables (as it does not have PK) CLOB columns appearing as OOS most of time Stores results / mismatching records in the DB Debugging issues is quite difficult
DB Solo	- Third party data comparison tool, capable of comparing heterogeneous databases. - Comes with UI and Linux based installation - Generates OOS sync report giving the mismatch/missing records. - Generates everything on the DB solo server machine and only reads from DB for data - Supports disk based and Memory based comparison - User has good control and can tweak parameters as per the needs - No separate DB setup is needed - This is scaling for PSP needs in pre-prod, still to bench mark it with Veridata numbers	- Very limited documentation available, only way to troubleshoot is to reach the support of DB Solo Doesn't generate repair scripts, in case of data modification it has to be done manually. It can generate repair script, generated repair script from the tool. - generates physical file when working on a disk based compare so we have to configure good amount of space on the server - Limitation with Memory based compare is we can compare only the data of memory size not more than that

DB SOLO

Table Data Comparison Tool allows you to compare data in one or more pairs of tables. The tables can be in the same or different schemas/databases and they don't need to have the same name or columns. Comparison can be done between tables in different DBMS platforms such as Oracle, SQL Server and PostgreSQL.

High Level Design Decisions:

1. Memory based comparison Vs Disk Based comparison

- We have decided to use Memory/ Disk based comparison as per the scenario
 - For QBDT_REQUEST_INFO and SAP_METHOD_CALL we have used Disk based comparison as the number of records were high and DB solo was taking time loading the records in memory
 - For SOURCE_SYSTEM_TRANSMISSION we have used memory based comparison as it was giving 30-40 % performance improvement

2. Partitioning

- We have used partitioning as per the data volume and number of records
 - For SOURCE_SYSTEM_TRANSMISSION we have used following partitioning
 - For 2009 - 2014 : We have used monthly Partitioning (Took varying time depending on data size from 20 mins - 1 hour)
 - For 2015 - 2022 : We have used 10 days partitions (On an average took 1 hour to compare one partition)
 - For QBDT_REQUEST_INFO we have used yearly partitions (nearly took 3 hours to compare 1 partition)
 - For SAP_METHOD_CALL we have used Monthly partitions (It is taking 45 mins - 1 hour) as Yearly partition was taking huge time (8 hours approx.)

3. Parallelism

- We have used parallelism using 4 machines
- This has benefitted us comparing 4 years data in 36 hours
- We can add more machines if we want to improve further on time (in case we need to do historical comparison again)

XML generated from DB solo for Linux comparison

```
<?xml version = '1.0' encoding = 'UTF-8'?>
<compare type="dataCompare" version="5.3.0">
```

Connection details

```
<source name="PostgreS_IBOB_Prd" id="1647240807664" schema="ibobadm" database="prodapgib"/>
<destination name="Oracle_ibob_prd_target" id="1655920776196" schema="IBOBADM" database=""/>
```

Table Details

```

<tables>
<gg_heartbeat included="false"/> -Whe
<psp_hcm401k_company_policy includ
<psp_hcm401k_company_qbdt_pitem i
.....
<psp_hcm401k_employee_deduction included="false"/>
<psp_hcm401k_policy included="false"/>
<psp_qbdt_request_info included="true" where="" mappedwhere="" pk_order="" mapped="PSP_QBDT_REQUEST_INFO" mapped_pk_order=""> - When included is marked as true DB solo will consider that table for comparison / Mapped
points the table name from target and the starting tag gives the table name from source similar thing follows in columns
<columns>
<qbdt_request_info_seq included="true" pk="true" function="" mappedfunction="" mapped="QBDT_REQUEST_INFO_SEQ" mappedpk="true"/> - Column level comparison can be controlled by changing Included field / Where Pk is true
it points the primary key column
<version included="true" pk="false" function="" mappedfunction="" mapped="VERSION" mappedpk="false"/>
<creator_id included="true" pk="false" function="" mappedfunction="" mapped="CREATOR_ID" mappedpk="false"/>
<created_date included="true" pk="false" function="" mappedfunction="" mapped="CREATED_DATE" mappedpk="false"/>
<modifier_id included="true" pk="false" function="" mappedfunction="" mapped="MODIFIER_ID" mappedpk="false"/>
<modified_date included="true" pk="false" function="" mappedfunction="" mapped="MODIFIED_DATE" mappedpk="false"/>
<realm_id included="true" pk="true" function="" mappedfunction="" mapped="REALM_ID" mappedpk="true"/>
<employee_add_count included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_ADD_COUNT" mappedpk="false"/>
<employee_add_start included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_ADD_START" mappedpk="false"/>
<employee_add_end included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_ADD_END" mappedpk="false"/>
<employee_update_count included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_UPDATE_COUNT" mappedpk="false"/>
<employee_update_start included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_UPDATE_START" mappedpk="false"/>
<employee_update_end included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_UPDATE_END" mappedpk="false"/>
<paycheck_add_count included="true" pk="false" function="" mappedfunction="" mapped="PAYCHECK_ADD_COUNT" mappedpk="false"/>
<paycheck_update_count included="true" pk="false" function="" mappedfunction="" mapped="PAYCHECK_UPDATE_COUNT" mappedpk="false"/>
<payroll_processing_start included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_PROCESSING_START" mappedpk="false"/>
<payroll_processing_end included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_PROCESSING_END" mappedpk="false"/>
<payroll_item_add_count included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_ADD_COUNT" mappedpk="false"/>
<payroll_item_add_start included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_ADD_START" mappedpk="false"/>
<payroll_item_add_end included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_ADD_END" mappedpk="false"/>
<payroll_item_update_count included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_UPDATE_COUNT" mappedpk="false"/>
<payroll_item_update_start included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_UPDATE_START" mappedpk="false"/>
<payroll_item_update_end included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_UPDATE_END" mappedpk="false"/>
<payroll_transaction_add_count included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANSACTION_ADD_COUNT" mappedpk="false"/>
<payroll_transaction_add_start included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANSACTION_ADD_START" mappedpk="false"/>
<payroll_transaction_add_end included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANSACTION_ADD_END" mappedpk="false"/>
<payroll_trans_update_count included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANS_UPDATE_COUNT" mappedpk="false"/>
<payroll_trans_update_start included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANS_UPDATE_START" mappedpk="false"/>
<payroll_trans_update_end included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANS_UPDATE_END" mappedpk="false"/>
<employee_delete_count included="true" pk="false" function="" mappedfunction="" mapped="EMPLOYEE_DELETE_COUNT" mappedpk="false"/>
<payroll_item_delete_count included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_ITEM_DELETE_COUNT" mappedpk="false"/>
<payroll_trans_delete_count included="true" pk="false" function="" mappedfunction="" mapped="PAYROLL_TRANS_DELETE_COUNT" mappedpk="false"/>
<delete_processing_start included="true" pk="false" function="" mappedfunction="" mapped="DELETE_PROCESSING_START" mappedpk="false"/>
<delete_processing_end included="true" pk="false" function="" mappedfunction="" mapped="DELETE_PROCESSING_END" mappedpk="false"/>
<source_system_transmission_fk included="true" pk="false" function="" mappedfunction="" mapped="SOURCE_SYSTEM_TRANSMISSION_FK" mappedpk="false"/>
<paycheck_delete_count included="true" pk="false" function="" mappedfunction="" mapped="PAYCHECK_DELETE_COUNT" mappedpk="false"/>
</columns>
</psp_qbdt_request_info>
<psp_sap_method_call included="true" where="" mappedwhere="" pk_order="" mapped="PSP_SAP_METHOD_CALL" mapped_pk_order="">
<columns>
<sap_method_call_seq included="true" pk="true" function="" mappedfunction="" mapped="SAP_METHOD_CALL_SEQ" mappedpk="true"/>
<version included="true" pk="false" function="" mappedfunction="" mapped="VERSION" mappedpk="false"/>
<creator_id included="true" pk="false" function="" mappedfunction="" mapped="CREATOR_ID" mappedpk="false"/>
<created_date included="true" pk="false" function="" mappedfunction="" mapped="CREATED_DATE" mappedpk="false"/>
<modifier_id included="true" pk="false" function="" mappedfunction="" mapped="MODIFIER_ID" mappedpk="false"/>
<modified_date included="true" pk="false" function="" mappedfunction="" mapped="MODIFIED_DATE" mappedpk="false"/>
<realm_id included="true" pk="true" function="" mappedfunction="" mapped="REALM_ID" mappedpk="true"/>
<screen_path included="true" pk="false" function="" mappedfunction="" mapped="SCREEN_PATH" mappedpk="false"/>
<service_name included="true" pk="false" function="" mappedfunction="" mapped="SERVICE_NAME" mappedpk="false"/>
<method_name included="true" pk="false" function="" mappedfunction="" mapped="METHOD_NAME" mappedpk="false"/>
<parameters included="true" pk="false" function="" mappedfunction="" mapped="PARAMETERS" mappedpk="false"/>
<result_size included="true" pk="false" function="" mappedfunction="" mapped="RESULT_SIZE" mappedpk="false"/>
<elapsed_millis included="true" pk="false" function="" mappedfunction="" mapped="ELAPSED_MILLIS" mappedpk="false"/>
<exception_trace included="true" pk="false" function="" mappedfunction="" mapped="EXCEPTION_TRACE" mappedpk="false"/>
<security_principal included="true" pk="false" function="" mappedfunction="" mapped="SECURITY_PRINCIPAL" mappedpk="false"/>
<session_id included="true" pk="false" function="" mappedfunction="" mapped="SESSION_ID" mappedpk="false"/>
<host included="true" pk="false" function="" mappedfunction="" mapped="HOST" mappedpk="false"/>

```

```

</columns>
</psp_sap_method_call>
<psp_source_system_transmission included="true" where="created_date between '2000-08-01 00:00:00' and '2000-08-31 23:59:59' mappedwhere="(created_date between to_date('2000-08-01 00:00:00','yyyy-mm-dd hh24:mi:ss') and
to_date('2000-08-31 23:59:59','yyyy-mm-dd
<columns>
<source_system_transmission_seq included="true" pk="true" function="" mappedfunction="" mapped="SOURCE_SYSTEM_TRANSMISSION_SEQ" mappedpk="true"/>
<version included="true" pk="false" function="" mappedfunction="" mapped="VERSION" mappedpk="false"/>
<creator_id included="true" pk="false" function="" mappedfunction="" mapped="CREATOR_ID" mappedpk="false"/>
<created_date included="true" pk="false" function="" mappedfunction="" mapped="CREATED_DATE" mappedpk="false"/>
<modifier_id included="true" pk="false" function="" mappedfunction="" mapped="MODIFIER_ID" mappedpk="false"/>
<modified_date included="true" pk="false" function="" mappedfunction="" mapped="MODIFIED_DATE" mappedpk="false"/>
<realm_id included="true" pk="true" function="" mappedfunction="" mapped="REALM_ID" mappedpk="true"/>
<from_source_system included="true" pk="false" function="" mappedfunction="" mapped="FROM_SOURCE_SYSTEM" mappedpk="false"/>
<request_token included="true" pk="false" function="" mappedfunction="" mapped="REQUEST_TOKEN" mappedpk="false"/>
<response_token included="true" pk="false" function="" mappedfunction="" mapped="RESPONSE_TOKEN" mappedpk="false"/>
<request_document_old included="true" pk="false" function="" mappedfunction="" mapped="REQUEST_DOCUMENT_OLD" mappedpk="false"/>
<response_document_old included="true" pk="false" function="" mappedfunction="" mapped="RESPONSE_DOCUMENT_OLD" mappedpk="false"/>
<type included="true" pk="false" function="" mappedfunction="" mapped="TYPE" mappedpk="false"/>
<initialize_date_time included="true" pk="false" function="" mappedfunction="" mapped="INITIALIZE_DATE_TIME" mappedpk="false"/>
<finalize_date_time included="true" pk="false" function="" mappedfunction="" mapped="FINALIZE_DATE_TIME" mappedpk="false"/>
<description included="true" pk="false" function="" mappedfunction="" mapped="DESCRIPTION" mappedpk="false"/>
<to_source_system included="true" pk="false" function="" mappedfunction="" mapped="TO_SOURCE_SYSTEM" mappedpk="false"/>
<transmission_identifier included="true" pk="false" function="" mappedfunction="" mapped="TRANSMISSION_IDENTIFIER" mappedpk="false"/>
<i_p_address included="true" pk="false" function="" mappedfunction="" mapped="I_P_ADDRESS" mappedpk="false"/>
<company_id included="true" pk="false" function="" mappedfunction="" mapped="COMPANY_ID" mappedpk="false"/>
<host included="true" pk="false" function="" mappedfunction="" mapped="HOST" mappedpk="false"/>
<application_version included="true" pk="false" function="" mappedfunction="" mapped="APPLICATION_VERSION" mappedpk="false"/>
<application_id included="true" pk="false" function="" mappedfunction="" mapped="APPLICATION_ID" mappedpk="false"/>
<tax_table_id included="true" pk="false" function="" mappedfunction="" mapped="TAX_TABLE_ID" mappedpk="false"/>
<response_document included="true" pk="false" function="" mappedfunction="case when dbms_lob.getlength(#COLUMN#) =0 then null else #COLUMN# end" mapped="RESPONSE_DOCUMENT" mappedpk="false"/> - Mapped function
will contain the function for transformation at target and Function field will contain the same for source
<request_document included="true" pk="false" function="" mappedfunction="case when dbms_lob.getlength(#COLUMN#) =0 then null else #COLUMN# end" mapped="REQUEST_DOCUMENT" mappedpk="false"/>
</columns>
</psp_source_system_transmission>
</tables>
</settings>

```

Config parameters / settings

```

<datacomp_emptystrings_as_nulls value="false"/> - Compare empty strings as nulls when set to true (recommendation "false")
<datacomp_ignorestringcase value="false"/> - Ignores string case when set to true (recommendation "false")
<datacomp_trimstring value="false"/> - Trims the string when set to true (recommendation "false")
<datacomp_ignorenumpersign value="false"/> - Ignores number sign (recommendation "false")
<datacomp_decimal_scale value="-1"/> - Self explanatory
<datacomp_comparenumberdecimals value="true"/> - Self explanatory
<datacomp_comparedateyear value="true"/> - Self explanatory
<datacomp_comparedatemonth value="true"/> - Self explanatory
<datacomp_comparedateday value="true"/> - Self explanatory
<datacomp_comparedatehour value="true"/> - Self explanatory
<datacomp_comparedateminute value="true"/> - Self explanatory
<datacomp_comparedatesecond value="true"/> - Self explanatory
<datacomp_comparedatemillisecond value="true"/> - Self explanatory
<datacomp_ignorematching value="true"/> - Ignores matching values from the report (recommendation "true")
<datacomp_maxrows value="1000000"/> - Max number of records compared in a single run (recommendation : based on your requirement )
<datacomp_filebased value="false"/> - When "false" performs memory based comparison when "true" performs disk based comparison ( recommendation : based on your requirement )
<datacomp_temp_dir value="/Users/skumar71/Documents/PSP"/> - Path where temporary data files will be generated for comparison in case of disk based comparison. Disk based comparison will fail if left blank
<datacomp_rows_per_tempfile value="16384"/> - Internal to DB solo no need to change
<datacomp_rows_per_tempfile_blob value="1024"/> - Internal to DB solo no need to change
<datacomp_missing_table_error value="true"/> - To report missing table at target
<datacomp_missing_column_error value="true"/> - to report missing column at target
<datacomp_extra_table_include value="false"/> - To report extra table at target
<datacomp_extra_column_include value="true"/> - To report extra column at target
<datacomp_sync_syncsource value="false"/> - If you want to sync source with the SQL generated (recommendation : "false" , go through the sql generated and then if required do it manually )
<datacomp_sync_upperkeywords value="true"/> - Parameters for sync.
<datacomp_sync_sepsemicolon value="true"/> - Parameters for sync.
<datacomp_sync_ident_brackets value="false"/> - Parameters for sync.
<datacomp_sync_ident_quotes value="false"/> - Parameters for sync.
<datacomp_sync_ident_custom value=""/> - Parameters for sync.

```

```
<datacomp_sync_commit value="0"/> - Parameters for sync.
<datacomp_sync_comments value="true"/> - Parameters for sync.
<datacomp_sync_schema value="true"/> - Parameters for sync.
<datacomp_sync_indent value="true"/>
<datacomp_sync_filename value="Com
<datacomp_sync_file_encoding value="UTF-8"/> - File Encoding
<datacomp_sync_directoryname value="/Users/skumar71/Documents/PSP/DbSolo"/> - Directory where the sync sql will be generated
<datacomp_loader_directory value="/Users/skumar71/Documents/PSP/DbSolo"/> - should be same as above (for generating SQL loader format of file )
<datacomp_sync_loader value="true"/> - Parameters for sync.
<datacomp_sync_disabletriggers value="false"/> - Parameters for sync.
<datacomp_sync_disablefk value="false"/> - Parameters for sync.
<datacomp_sync_delete value="true"/> - Parameters for sync.
<datacomp_sync_identity_on value="true"/> - Parameters for sync.
<datacomp_sync_update value="true"/> - Parameters for sync.
<datacomp_sync_insert value="true"/> - Parameters for sync.
<datacomp_sync_directory value="true"/> - Parameters for sync ( this parameter defines whether to generate sync file or not )
<datacomp_html_file value="/Users/skumar71/Documents/PSP/DbSolo/db_solo-master 2/PSP_DBSOLO/Reports/source_system_memory_based_90days.html"/> - Comparison file report file path, file generation will fail if path is wrong
<datacomp_html_title value="DB Solo - Data Compare Results"/> - Self explanatory
<datacomp_html_heading value="DB Solo - Data Compare Results"/> - Self explanatory
<datacomp_html_header_background value="209,209,209"/> - Report alignment parameters
<datacomp_html_header_fontsize value="11"/> - Report alignment parameters
<datacomp_html_header_alignment value="1"/> - Report alignment parameters
<datacomp_html_even_background value="238,238,238"/> - Report alignment parameters
<datacomp_html_even_fontsize value="11"/> - Report alignment parameters
<datacomp_html_even_alignment value="1"/> - Report alignment parameters
<datacomp_html_odd_background value="221,221,221"/> - Report alignment parameters
<datacomp_html_odd_fontsize value="11"/> - Report alignment parameters
<datacomp_html_odd_alignment value="1"/> - Report alignment parameters
<datacomp_html_diff_background value="239,242,202"/> - Report alignment parameters
<datacomp_html_diff_fontsize value="11"/> - Report alignment parameters
<datacomp_html_diff_alignment value="1"/> - Report alignment parameters
<datacomp_html_details value="true"/> - Report contents parameter
<datacomp_html_details_include_matching value="false"/> - Report contents parameter
<datacomp_html_details_timestamp value="false"/> - Report contents parameter
<datacomp_html_include_matching value="true"/> - Report contents parameter
</settings>
</compare>
```

Comparison Summary Pre-Prod

Table Name	No. Of records	size	Duration	Type of comparison	Start time	End time	Elapsed Time
PSP_QBDT_REQUEST_INFO	195825	0.15 GB	Full Data	Memory Based	2022-02-03 08:24:12	2022-02-03 08:24:32	20 secs
PSP_QBDT_REQUEST_INFO	195825	0.15 GB	Full Data	Disk Based	2022-02-03 08:29:21	2022-02-03 08:30:50	1 min 29 secs
PSP_SAP_METHOD_CALL	1973147	1.26 GB	Full Data	Memory Based	2022-02-03 08:47:40	2022-02-03 08:51:28	3 mins 48 secs
PSP_SAP_METHOD_CALL	1973147	1.26 GB	Full Data	Disk Based	2022-02-03 08:35:10	2022-02-03 08:44:41	9 mins 31 secs
PSP_SOURCE_SYSTEM_TRANSMISSION	435525	2.2 GB	10 days	Memory Based	2022-02-03 09:12:20	2022-02-03 09:28:40	15 mins 40 secs
PSP_SOURCE_SYSTEM_TRANSMISSION	435525	2.2 GB	10 days	Disk Based	2022-02-03 10:05:03	2022-02-03 10:22:00	16 mins 57 secs
PSP_SOURCE_SYSTEM_TRANSMISSION	3245706	16 GB	90 days	Memory Based	2022-02-03 23:14:13	2022-02-04 00:16:17	1 hour 2 mins 4 secs
PSP_SOURCE_SYSTEM_TRANSMISSION	3245706	16 GB	90 days	Disk Based	2022-02-03 11:19:13	2022-02-03 11:57:19	38 mins 6 secs

Comparison Pre-Prod (Source System Transmission)

Source : Aurora PostgreSQL || Target : Oracle

Type of Comparison: Disk Based (We have decided to move forward with Disk based comparison as we can't scale the memory to PSP needs)

Mismatch Report Generated : Yes (We have simulated the mismatch by removing the matching function)

In Oracle if there is no value in Clob column it is considered as empty but in PostgreSQL same value is considered as NULL.

Table name	No. of Records	Duration	Start time	End Time	Elapsed Time (h:mm:ss)	#Mismatch
PSP_SOURCE_SYSTEM_TRANSMISSION						
PSP_SOURCE_SYSTEM_TRANSMISSION	1317487	2011 - 2014	2022-03-21 11:06:40	2022-03-21 11:43:52	0:37:12	60415
PSP_SOURCE_SYSTEM_TRANSMISSION	379860	2015 - 2018	2022-03-21 11:44:23	2022-03-21 11:58:41	0:14:18	131
PSP_SOURCE_SYSTEM_TRANSMISSION	3096218	2019 (9 months)	2022-03-21 11:58:45	2022-03-21 12:59:49	1:01:04	31
PSP_SOURCE_SYSTEM_TRANSMISSION	3443939	2019 (October)	2022-03-21 12:59:55	2022-03-21 14:05:05	1:05:10	11
PSP_SOURCE_SYSTEM_TRANSMISSION	2176456	2019 (November)	2022-03-21 14:05:11	2022-03-21 14:45:53	0:40:42	11
PSP_SOURCE_SYSTEM_TRANSMISSION	2015080	2019 (December)	2022-03-21 14:45:57	2022-03-21 15:24:59	0:39:02	8
PSP_SOURCE_SYSTEM_TRANSMISSION	3024920	2020 (January)	2022-03-21 15:25:03	2022-03-21 16:28:18	1:03:15	5
PSP_SOURCE_SYSTEM_TRANSMISSION	2593218	2020 (February)	2022-03-21 16:28:24	2022-03-21 17:17:53	0:49:29	8
PSP_SOURCE_SYSTEM_TRANSMISSION	2682951	2020 (March)	2022-03-21 17:17:59	2022-03-21 18:06:23	0:48:24	14
PSP_SOURCE_SYSTEM_TRANSMISSION	2981705	2020 (April)	2022-03-21 18:06:29	2022-03-21 19:02:51	0:56:22	58
PSP_SOURCE_SYSTEM_TRANSMISSION	2847456	2020 (May)	2022-03-21 19:02:57	2022-03-21 19:55:24	0:52:27	8
PSP_SOURCE_SYSTEM_TRANSMISSION	2109513	2020 (June- Dec)	2022-03-21 19:55:30	2022-03-21 20:40:54	0:45:24	492
PSP_SOURCE_SYSTEM_TRANSMISSION	2452629	2021	2022-03-21 20:41:00	2022-03-21 21:06:13	0:25:13	22
PSP_SOURCE_SYSTEM_TRANSMISSION	3173183	2022	2022-03-21 21:06:16	2022-03-21 21:35:32	0:29:16	9

Reports Generated from DB solo

- 1. Data compare report : This report indicates over stats like # of records compared, # of records missing, extra or Different.

DR Solo - Data Compare Results

Summary

Report Generated : **Mon Mar 21 21:35:32 PDT 2022**
Source Database : [PostgreSQL_IBOB_PRPRD] [pspsys01] [ibobadm]
Destination Database : [Oracle_IBOB_PRPRD] [IBOBADM_APG_CLOB_PATCH]
Number of compared tables/views : **1**
Number of non-matching tables/views : **1**
Comparison Started : **2022-03-21 21:06:16**
Comparison Ended : **2022-03-21 21:35:32**

Results

Name	Matching	Different	Missing	Extra
psp_source_system_transmission	3173183	2	0	0

Mon Mar 21 21:35:32 PDT 2022

Generated Using [DB Solo](#)

2. Mismatch : This kind of report is generated when there is a difference in source and target, Columns having difference in values are highlighted as shown below.
- a. In this report the first 2 columns specify the primary key followed by the columns from Source and Target (First Source column value will be printed followed by Same target column)

psp_source_system_transmission

Different

source_system_transmission_seq	realm_id	 version	 VERSION	 creator_id	 CREATOR_ID	 created_date	 CREATED_DATE	 modifier_id	 MODIFIER_ID	 modified_date	 MODIFIED_DATE	 from_source_system	 FROM_SOURCE_SYSTEM	 request_token	 REQUEST_TOKEN	 response_token	 RESPONSE_TOKEN	 type	
0153ca8a-2a72-4c79-8926-b0426af9140b	-1	1	1	ERSBatchJob	ERSBatchJob	2022-01-07 19:00:18.262	2022-01-07 19:00:18.262	ERSBatchJob	ERSBatchJob	2022-01-07 19:00:18.262	2022-01-07 19:00:18.262	PSP	PSP	0	0	0	0	EntitlementUnitActivation	En
0e872c8d-52ec-4571-a632-318532fdded53	-1	1	1	ERSBatchJob	ERSBatchJob	2022-01-14 19:00:20.939	2022-01-14 19:00:20.939	ERSBatchJob	ERSBatchJob	2022-01-14 19:00:20.94	2022-01-14 19:00:20.94	PSP	PSP	0	0	0	0	EntitlementUnitActivation	En
437387a3-b3d8-4d51-a712-d0e5c5031897	-1	1	1	ERSBatchJob	ERSBatchJob	2022-01-10 19:00:15.447	2022-01-10 19:00:15.447	ERSBatchJob	ERSBatchJob	2022-01-10 19:00:15.447	2022-01-10 19:00:15.447	PSP	PSP	0	0	0	0	EntitlementUnitActivation	En
58edff9f-f242-429f-af70-13f200f14da5	-1	1	1	ERSBatchJob	ERSBatchJob	2022-02-23 16:52:00.211	2022-02-23 16:52:00.211	ERSBatchJob	ERSBatchJob	2022-02-23 16:52:00.211	2022-02-23 16:52:00.211	PSP	PSP	0	0	0	0	EntitlementUnitActivation	En

[illegible]

3. Extra : This report is generated when the records are present in source but not in Target:

PSP_QBDT_REQUEST_INFO

Extra

QBDT_REQUEST_INFO_SEQ	VERSION	CREATOR_ID	CREATED_DATE	MODIFIER_ID	MODIFIED_DATE	REALM_ID	EMPLOYEE_ADD_COUNT	EMPLOYEE_ADD_START	EMPLOYEE_ADD_END	EMPLOYEE_UPDATE_COUNT	EMPLOYEE_UPDATE_START	EMPLOYEE_UPDATE_END	FATCHECK_ADD_COUNT	FATCHECK_ADD_START	FATCHECK_ADD_END
00e6805a-f4b1-41ea-9cf0-87a223ce55fd	0	QBDTAdapter	2022-03-13 03:16:06.031	QBDTAdapter	2022-03-13 03:16:06.031	-1	0	2022-03-13 03:16:04.131	2022-03-13 03:16:04.131	5	2022-03-13 03:16:04.131	2022-03-13 03:16:04.338	5		0
00f4c118-42ef-4494-b5eb-738be9c1f14b	0	QBDTAdapter	2022-03-13 03:19:01.999	QBDTAdapter	2022-03-13 03:19:01.999	-1	0	2022-03-13 03:19:00.393	2022-03-13 03:19:00.393	1	2022-03-13 03:19:00.393	2022-03-13 03:19:00.485	1		0
017ac23a-8682-4a9b-ac6b-f373769c21fe	0	QBDTAdapter	2022-03-13 03:03:19.038	QBDTAdapter	2022-03-13 03:03:19.038	-1	0	2022-03-13 03:03:17.632	2022-03-13 03:03:17.632	8	2022-03-13 03:03:17.632	2022-03-13 03:03:17.74	8		0
039cdde3-168d-4a33-91e3-6a0116e50e12	0	QBDTAdapter	2022-03-13 03:19:09.424	QBDTAdapter	2022-03-13 03:19:09.424	-1	0	2022-03-13 03:19:07.838	2022-03-13 03:19:07.838	6	2022-03-13 03:19:07.838	2022-03-13 03:19:08.028	6		2
09160577-f975-4f1e-bdcf-e6cab8931ad7	0	QBDTAdapter	2022-03-13 03:14:24.894	QBDTAdapter	2022-03-13 03:14:24.894	-1	0	2022-03-13 03:14:24.737	2022-03-13 03:14:24.737	1	2022-03-13 03:14:24.737	2022-03-13 03:14:24.746	1		0
0a3966ca-f673-4eb3-bdfa-5f70a3ab5277	0	QBDTAdapter	2022-03-13 03:10:35.919	QBDTAdapter	2022-03-13 03:10:35.919	-1	0	2022-03-13 03:10:35.812	2022-03-13 03:10:35.812	1	2022-03-13 03:10:35.812	2022-03-13 03:10:35.833	1		0
0ca7f980-5409-45b1-a710-99ee92a4954b	0	QBDTAdapter	2022-03-13 03:01:08.902	QBDTAdapter	2022-03-13 03:01:08.902	-1	0	2022-03-13 03:01:05.289	2022-03-13 03:01:05.289	8	2022-03-13 03:01:05.289	2022-03-13 03:01:05.671	8		0
0d844460-f676-4168-bf66-2fbcc956db66	0	QBDTAdapter	2022-03-13 03:06:56.847	QBDTAdapter	2022-03-13 03:06:56.847	-1	0	2022-03-13 03:06:55.62	2022-03-13 03:06:55.622	32	2022-03-13 03:06:55.622	2022-03-13 03:06:56.065	31		0
10fb0caf-02fa-4268-9ae0-81a80a4d988a	0	QBDTAdapter	2022-03-14 09:20:26.529	QBDTAdapter	2022-03-14 09:20:26.529	-1	0	2022-03-14 09:20:24.296	2022-03-14 09:20:24.296	10	2022-03-14 09:20:24.296	2022-03-14 09:20:24.397	1		60
12ae38fe-cfbd-422c-bda2-4561e125efe1	0	QBDTAdapter	2022-03-13 03:15:17.001	QBDTAdapter	2022-03-13 03:15:17.001	-1	0	2022-03-13 03:15:16.162	2022-03-13 03:15:16.162	3	2022-03-13 03:15:16.162	2022-03-13 03:15:16.335	3		0
14da9e8a-e79d-436a-8ae3-ee3f981a9630	0	QBDTAdapter	2022-03-13 03:08:59.261	QBDTAdapter	2022-03-13 03:08:59.261	-1	0	2022-03-13 03:08:57.844	2022-03-13 03:08:57.844	2	2022-03-13 03:08:57.844	2022-03-13 03:08:57.915	2		0
17a95737-6aa8-4800-85a0-279353b41676	0	QBDTAdapter	2022-03-13 03:03:25.494	QBDTAdapter	2022-03-13 03:03:25.494	-1	0	2022-03-13 03:03:25.033	2022-03-13 03:03:25.033	3	2022-03-13 03:03:25.033	2022-03-13 03:03:25.088	3		0
25920879-9a15-4c95-a76b-5d4f576b3355	0	QBDTAdapter	2022-03-13 03:13:36.841	QBDTAdapter	2022-03-13 03:13:36.841	-1	0	2022-03-13 03:13:36.774	2022-03-13 03:13:36.776	1	2022-03-13 03:13:36.776	2022-03-13 03:13:36.797	0		0
26c2a325-45c0-4a5f-83cf-1795ad592a32	0	QBDTAdapter	2022-03-13 03:18:22.898	QBDTAdapter	2022-03-13 03:18:22.898	-1	0	2022-03-13 03:18:22.755	2022-03-13 03:18:22.755	5	2022-03-13 03:18:22.755	2022-03-13 03:18:22.775	5		0
2f96ccf5-6876-4b75-86ed-964f01115d8d	0	QBDTAdapter	2022-03-13 03:11:51.732	QBDTAdapter	2022-03-13 03:11:51.732	-1	0	2022-03-13 03:11:51.634	2022-03-13 03:11:51.635	1	2022-03-13 03:11:51.635	2022-03-13 03:11:51.663	0		0
2fec6bud-8bd5-485c-b0d2-9df68eb0447b	0	QBDTAdapter	2022-03-13 03:01:02.929	QBDTAdapter	2022-03-13 03:01:02.929	-1	0	2022-03-13 03:01:01.23	2022-03-13 03:01:01.23	4	2022-03-13 03:01:01.23	2022-03-13 03:01:01.389	4		0
322756c5-e32a-488b-b672-4bdd2a72079	0	QBDTAdapter	2022-03-13 03:15:25.457	QBDTAdapter	2022-03-13 03:15:25.457	-1	0	2022-03-13 03:15:24.599	2022-03-13 03:15:24.599	2	2022-03-13 03:15:24.599	2022-03-13 03:15:24.691	2		0
3617fc4a-a097-4d0c-9303-91a691926b19	0	QBDTAdapter	2022-03-13 03:01:13.884	QBDTAdapter	2022-03-13 03:01:13.884	-1	0	2022-03-13 03:01:13.659	2022-03-13 03:01:13.659	3	2022-03-13 03:01:13.659	2022-03-13 03:01:13.698	3		0
3840c24a-bfa0-44c8-aa13-070a724d6c8a	0	QBDTAdapter	2022-03-13 03:07:53.882	QBDTAdapter	2022-03-13 03:07:53.882	-1	0	2022-03-13 03:07:53.58	2022-03-13 03:07:53.58	2	2022-03-13 03:07:53.58	2022-03-13 03:07:53.618	2		0

4. Missing : This report is generated when the records are deleted from source but still present in Target:

PSP_SAP_METHOD_CALL

Missing

sap_method_call_seq	version	creator_id	created_date	modifier_id	modified_date	realm_id	screen_path	service_name	method_name	parameters	result_size	elapsed_millis	exception_trace	security_principal	session_id	host
01a1043f-a54f-4015-aacc-2368947dcd54	1	50002335694	2022-03-13 01:56:17.689	50002335694	2022-03-13 01:56:17.714	-1	Company	payrollservice	getTransactionTypeList	NULL	150	10	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-zjbqf
43e12be2-0589-490e-8b04-028822738d3e	1	50002335694	2022-03-13 01:56:17.458	50002335694	2022-03-13 01:56:17.478	-1	Company	companyservice	getFundingModelList	NULL	3	4	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-bfmt5
4c9dc1aa-5c98-4033-9701-67272d7acdeb	1	50002335694	2022-03-13 01:56:17.566	50002335694	2022-03-13 01:56:17.58	-1	Company	companyservice	getSourceSystemList	NULL	9	1	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-8wxlp
5b00074d-66c9-4f29-93b9-9795fc9efcdb	1	50002335694	2022-03-13 01:56:17.411	50002335694	2022-03-13 01:56:17.429	-1	Company	companyservice	getServiceStatusList	NULL	29	4	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-bddrv
5eb73a55-f3e0-4720-9200-fbb108e97862	1	50002335694	2022-03-13 01:56:17.537	50002335694	2022-03-13 01:56:17.557	-1	Company	companyservice	getServiceSubStatusList	NULL	29	5	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-8lrbg
917a02c8-b203-415f-b246-862aeca7dae5	1	50002335694	2022-03-13 01:56:17.686	50002335694	2022-03-13 01:56:17.88	-1	Company	taxservice	getAgencyList	NULL	91	177	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-xkw4r
f020521a-ae85-438a-8f5c-7c9ef56e9185	1	50002335694	2022-03-13 01:56:17.908	50002335694	2022-03-13 01:56:17.921	-1	Company	companyservice	getFraudEventTypes	NULL	20	2	NULL	Paula Torres	04ba482b-86f0-4a50-8d8e-dd7123c7ae31	psp-sap-deployment-776c7759f5-dztdc

Repair Script : In case if there is any difference in source and target accordingly a repair sql script will be generated so that we can sync Source and Target.

Mismatch : will generate Update script

Missing : will generate delete script

Extra : will generate Insert script

For generating the repair script below parameter need to set to "True":

```
<datacomp_sync_directory value="true"/>
```

below is the place where you can define the location in which. you want to generate the script :

```
<datacomp_sync_directoryname value=""/>
```

1 Comment



Dubey, Vivek

To compare a VARCHAR column wil

```
<CLIENT_TXN_ID json_semantic="false" included="true" pk="false" function="" mappedfunction="case when length(#COLUMN#)=0 then null else #COLUMN# end" mapped="client_txn_id" mappedpk="false"/>
<RSS_REQUEST_ID json_semantic="false" included="true" pk="false" function="" mappedfunction="case when length(#COLUMN#)=0 then null else #COLUMN# end" mapped="rss_request_id" mappedpk="false"/>
```